

NAVEEN SHAJI GEORGE

Robotics Engineer | Autonomous Systems | Embedded Robotics | Robotics Hardware

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PROFESSIONAL SUMMARY

Robotics and Automation Engineer with hands-on experience in embedded systems, robotics hardware, PCB design, electrical systems, autonomous vehicles, and robot integration. Worked on autonomous vehicle and mobile robotics projects involving ESP32, Raspberry Pi 4, ROS 2, LiDAR, CAN communication, sensors, actuators, and custom PCBs. Served as Electrical & Electronics Head of Team Equinox for aBAJA SAEINDIA, contributing to electrical architecture, ECU development, wiring harnesses, drive-by-wire systems, and vehicle testing.

ENGINEERING EXPERIENCE

Team Equinox – Saintgits College of Engineering

2024 – 2026

Electrical & Electronics Head | aBAJA SAEINDIA

AIR 7 – 2025 | AIR 11 – 2026

- Worked on the electrical and electronic systems of an autonomous electric Baja vehicle, including ECUs, power distribution, CAN communication, sensors, actuators, and drive-by-wire systems.
- Designed custom front and rear ECU PCBs using KiCad, including schematics, component selection, PCB layout, power regulation, CAN interfaces, sensor interfaces, actuator interfaces, and board-level debugging.
- Developed and assembled LV/HV wiring harnesses for vehicle electrical systems, integrating batteries, DC-DC converters, motor controller, ECUs, sensors, actuators, and protection circuits.
- Worked on Throttle-by-Wire, Brake-by-Wire, and Steer-by-Wire systems using ESP32-based controllers, actuators, sensors, and CAN communication.
- Worked on the integration of the electrical systems with the vehicle's autonomous control architecture, including sensor interfaces and communication with the supervisory computing system.
- Led the Electrical & Electronics division, coordinating PCB development, wiring, component integration, testing, troubleshooting, technical documentation, and competition preparation.

Muthoot Hospitals, Kozhencherry

Jun 2025 – Jul 2025

Clinical Robotics Intern – Versius Surgical Robotic System

Medical Robotics

- Observed the operation and clinical workflow of the Versius surgical robotic system in a hospital environment.
- Gained exposure to robotic-assisted surgical procedures, robotic hardware, surgeon-console operation, system preparation, and clinical safety requirements.

SELECTED PROJECTS

Hospital Service AMR | Raspberry Pi 4, ROS 2, LiDAR, Nav2, Cartographer

2024 – 2025

- Developed a mobile hospital service robot for in-ward assistance and autonomous navigation.
- Used a Raspberry Pi 4 as the main computing platform and integrated LiDAR-based sensing with ROS 2 for robot navigation and mapping.
- Worked with the robot's omnidirectional drive platform, sensor integration, mapping, localization, and navigation.

RailGuard AI Anomaly Rover | Autonomous Rover, YOLOv8, Edge AI

2024

- Worked on an autonomous rover concept for railway inspection and anomaly detection.
- Integrated robotics hardware with computer-vision-based detection for identifying objects and anomalies during inspection.
- Participated in hardware development, robot integration, testing, and technical presentation of the system.

HeRo Swarm Robotics Study | ESP32, Gazebo, Multi-Agent Robotics

2024

- Studied multi-robot coordination using ESP32-based robot platforms and Gazebo simulation.
- Explored communication and consensus-based coordination approaches for multi-agent robotic systems.

Smart CCTV Edge Vision | C++, OpenCV, Intel oneAPI, YOLOv8

2024

- Developed a computer-vision-based surveillance system for real-time detection using edge computing.
- Worked with OpenCV, YOLOv8, C++, and Intel oneAPI for vision processing and inference.
- Selected among the Top 25 teams from 300+ teams in the Intel AI Hackathon.

Autonomous Baja Vehicle – Electrical System | KiCad, ESP32, CAN, PCB Design

2025 – 2026

- Designed and developed custom ECU hardware for vehicle control and monitoring applications.
- Worked with CAN communication, power regulation, sensor inputs, actuator control, watchdog systems, and vehicle safety circuits.
- Designed electrical architecture and wiring harnesses for integrating multiple ECUs and vehicle subsystems.

TECHNICAL SKILLS

Programming: C, C++, Python

Robotics: ROS 2, Gazebo, Nav2, Mobile Robotics, Autonomous Systems, Sensor Integration

Embedded Systems: ESP32, Arduino, Raspberry Pi 4, FreeRTOS, Embedded C/C++

Electronics: PCB Design, KiCad, Power Electronics, CAN, Sensors, Actuators, Motor Control, Wiring Harnesses

Communication Protocols: CAN, UART, I2C, SPI

Computer Vision: OpenCV, YOLOv8, Intel oneAPI

Engineering Tools: Git, GitHub, Linux, SolidWorks, Fusion 360, EasyEDA

EDUCATION

Saintgits College of Engineering

B.Tech. in Robotics & Automation

2022 – 2026

CGPA: 6.9 / 10

ACHIEVEMENTS

- **AIR 7 – aBAJA SAEINDIA 2025** with Team Equinox.
- **AIR 11 – aBAJA SAEINDIA 2026** with Team Equinox.
- **Top 25 – Intel AI Hackathon** among 300+ participating teams.
- **Electrical & Electronics Head** – Team Equinox, Saintgits College of Engineering.

LEADERSHIP & ACTIVITIES

- **Vice President – Anchoring Club**, Saintgits College of Engineering.
- **Coordinator – Campus Radio Jockey**, Saintgits College of Engineering.